

Part C

Course title: Bayes Methods
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1. In a study of the association of DNA mutations with hair color, DNA sequences were observed for n individuals and their hair color recorded. For $i = 1, 2, \dots, n$ let $y_i = 1$ if individual i has red hair and $y_i = 0$ if not, let $x_i = (x_{i,1}, \dots, x_{i,L})$ with $x_{i,j} = 0$ if there is no mutation at site $j = 1, 2, \dots, L$ and $x_{i,j} = 1$ if there is a mutation. Let $X = [x_{i,j}]_{i=1:n, j=1:L}$. The number of sites $L \gg 1$ is large.

Let α be an intercept parameter and $\beta = (\beta_1, \dots, \beta_L)$ a parameter vector of mutation effects. Let $z = (z_1, \dots, z_L)$ be a binary model indicator vector $z \in \Omega_z$ with $\Omega_z = \{0, 1\}^L$. Let $\beta_z = (z_1\beta_1, \dots, z_L\beta_L)^T$. Consider a logistic regression of y on X given z , $y_i \sim \text{Bernoulli}(p_i)$ with $p_i(\alpha, \beta, z) = \exp(\alpha + x_i\beta_z) / (1 + \exp(\alpha + x_i\beta_z))$.

(a) [10 marks]

- (i) Write down the posterior $\pi(\alpha, \beta, z|y)$ in as much detail as you can, but without eliciting a prior for $\pi(\alpha, \beta, z)$.

Solution: (Easy variation on bookwork)

$$L(\alpha, \beta, z; y) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

with $p_i(\alpha, \beta, z)$ given in the question. The posterior is

$$\pi(\alpha, \beta, z|y) = \frac{L(\alpha, \beta, z; y)\pi(\alpha, \beta, z)}{p(y)}$$

with marginal likelihood

$$p(y) = \sum_{z \in \Omega_z} p(y|z)\pi(z)$$

and marginal likelihood for model z

$$p(y|z) = \int L(\alpha, \beta, z; y)\pi(\beta|z)d\alpha d\beta$$

- (ii) It is of interest to see if any of the mutations are associated with hair color. If some of the mutations are associated with hair color then it is expected that the number will be small, perhaps four or five. Write down a prior $\pi(z)$ for z informed by this information.

Solution: (familiar type from classwork) The null model here is $z = (0, \dots, 0)$ corresponding to the hypothesis of no effects. If we assign this model prior probability one half, this is non-informative with respect to this hypothesis. Let $k = \sum_i z_i$. The z_i seem to be exchangeable, so we take

$$\pi(z|k > 0) = \frac{\xi^k (1 - \xi)^{L-k}}{1 - (1 - \xi)^L}.$$

Now $E(k|k > 0) = L\xi/[1 - (1 - \xi)^L]$ so if $E(k|k > 0) = k_0$ ($k_0 \simeq 4, 5$) then this fixes ξ and

$$\pi(z) = \frac{1}{2}\mathbb{I}_{k=0} + \frac{1}{2}\pi(z|k > 0).$$

Notice that ξ is $O(1/L)$ so $\pi(z)$ is $O(1/L^k)$

- (iii) You have access to a second large and completely independent data set of sequence data $\tilde{X} = [\tilde{x}]_{i=1:N, j=1:L}$ for individuals $i = 1, 2, \dots, N$ sampled at random from a large population the hair color is not recorded, but the proportion μ of red heads in the sampled population is known. Briefly, how does the data $\mu, \tilde{X}$ inform a prior for α, β given $\pi(z)$? How would you use simulation to validate the joint prior for α, β, z ?

Solution: (new) Our prior predictive distribution

$$p(y'|\tilde{X}) = \sum_z \int \prod_{i=1}^N \tilde{p}_i^{y'_i} (1 - \tilde{p}_i)^{1-y'_i} \pi(\alpha, \beta, z) d\alpha d\beta$$

with $\tilde{p}_i(\alpha, \beta, z) = \exp(\alpha + \tilde{x}_i \beta_z) / (1 + \exp(\alpha + \tilde{x}_i \beta_z))$ should predict a proportion $E(\bar{y}')$ approximately equal μ . This is easily verified by simulating $z \sim \pi(z)$, $\alpha, \beta \sim \pi(\alpha, \beta|z)$ and $y'_i \sim \text{Bernoulli}(\tilde{p}_i)$ for $i = 1, 2, \dots, N$ and comparing $\bar{y}'$ with μ . We could go further and determine a posterior for $\alpha, \beta, z|\tilde{X}, \mu$. Answers along these lines are fine, but hopefully discouraged by “briefly”.

- (b) [9 marks] Suppose $\pi(\alpha, \beta|z) = N(\alpha; 0, 1) \prod_{i=1}^n N(\beta_i; 0, 1)$ (ie, independent of z). Notice that we interpret $\beta_i z_i$ as the effect due to a mutation at site i , not β_i by itself.
- (i) Give a MCMC algorithm targeting $\pi(\beta, z|y)$, simplifying as far as possible.

Solution: (easy variation on classwork) We need updates for α, β and z . Treat α as β_0 . Here is something simple. The following applies for my choice of priors for $\beta|z$.

For each $i = 0, 1, \dots, L$, set $\beta' = \beta$ then replace $\beta'_i \sim N(0, 1)$ (ie, from its prior). The acceptance probability is

$$\alpha(\beta'|\beta) = \min \left\{ 1, \frac{L(\beta', z; y)}{L(\beta, z; y)} \right\}.$$

With probability $\alpha(\beta'|\beta)$ replace β' with β .

For each $i = 1, 2, \dots, L$, set $z' = z$ and replace $z'_i = 1 - z_i$. My priors for β don't depend on z so the acceptance probability is

$$\alpha(z'|z) = \min \left\{ 1, \frac{p_i^{y_i} (1 - p_i)^{1-y_i} \pi(z')}{p_i^{y_i} (1 - p_i)^{1-y_i} \pi(z)} \right\}$$

with $p'_i = p_i(\alpha, \beta, z')$. Here, $\pi(z')/\pi(z)$ could be simplified but this will depend on the prior choices above. With probability $\alpha(z'|z)$ replace z' with z .

- (ii) How would you test for significant effects? Is there a hazard for multiple testing in your approach? Justify your answer.

Solution: (new - just touched on in lectures) I would report the models z with larger values of the (estimated) posterior odds $\hat{O} = \hat{\pi}_z / \hat{\pi}_0$ ($z = 0$ is the model with no effects). The posterior probabilities (if they can be estimated with any precision) and the odds against the null do not suffer a hazard for multiple tests here because we condition on the data which is fixed. We have in a certain sense multiple data sets (the separate columns of X) and this would create a hazard, but this is met by our model prior which incorporates a Bonferroni style

correction to the posterior odds,

$$O = \frac{\pi(z|y)}{\pi(\underline{0}|y)} = \frac{p(y|z)\pi(z)}{p(y|\underline{0})\pi(\underline{0})}$$

which gives $O = B_{z,0}\pi(z|k > 0)$. Now $\pi(z|k > 0) \simeq \xi^k$ and ξ is $O(1/L)$ so $O = B_{z,0}O(1/L^k)$ so these odds downweight by the huge number of combinations of effects we are allowing. Bayes factors would be misleading.

(c) [6 marks]

- (i) The analysis is restricted to models $z \in A$ with $A \subseteq \Omega_z$. How would you modify your MCMC to simulate $\pi(z|z \in A, y)$? State any assumptions concerning A .

Solution: (variation on standard problem) Start the MCMC with $z \in A$ and reject (and update the step counter) if the proposal for z leaves A . We need A to be connected under the operations $z \rightarrow z'$ with $|z' - z| = 1$ so that the MCMC remains irreducible.

- (ii) Let B_A be the Bayes factor testing $z \in A$ against $z \in \Omega_z$. Show that

$$B_A = \frac{\pi(A|y)}{\pi(A)},$$

where $\pi(A|y) = \sum_{z \in A} \pi(z|y)$ and $\pi(A) = \sum_{z \in A} \pi(z)$ and state briefly how you would estimate this.

Solution: (new)

$$\begin{aligned} B_A &= \frac{p(y|z \in A)}{p(y)} \\ &= \frac{p(y, z \in A)}{p(z \in A)p(y)} \\ &= \frac{p(z \in A|y)}{\pi(A)} \\ &= \frac{p(z \in A|y)}{\pi(A)} \end{aligned}$$

2. For $i = 1, 2, \dots, n$ let $Y_i \sim f(y_i|x_i)$ when $X = x$ with $x = (x_1, \dots, x_n)$. Suppose $X \sim \pi(x|\theta)$ and $\theta \sim \pi(\theta)$ with θ a real scalar parameter.

(a) [15 marks] Suppose we observe $X = x$ and wish to estimate θ .

- (i) Explain what it means for $\pi(\theta|x)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $X = x$.

Solution: (bookwork) $\pi(\theta|x) \propto \pi(x|\theta)\pi(\theta)$. This is DI if either $\pi(x|\theta)$ or $\pi(\theta)$ are intractable functions of θ . This may happen if for example the normalisation of $\pi(x, \theta)$ is not available in closed form. MCMC targeting $\pi(\theta|x)$ does not require the overall normalisation $p(x)$ for $\pi(x|\theta)\pi(\theta)$ as it cancels in the Metropolis ratio $\pi(x|\theta')\pi(\theta')/\pi(x|\theta)\pi(\theta)$. However if $\pi(x|\theta)$ or $\pi(\theta)$ are intractable then this cannot be evaluated and so MH-MCMC is not possible.

- (ii) Let $S(x) = (S_1(x), \dots, S_p(x))$ be a p -component vector of statistics computed from the data. For $S, S' \in \mathbb{R}^p$ let $d(S, S')$ be a distance function and let $\rho > 0$ be a simulation parameter. Write down an ABC algorithm simulating

$$\theta \sim \pi(\theta|d(S(X), S(x)) < \rho).$$

and prove that your algorithm targets $\pi(\theta|d(S(X), S(x)) < \rho)$.

Solution: (bookwork)

(1) Simulate $\theta \sim \pi(\theta)$ and $x' \sim \pi(x'|\theta)$.

(2) If $d(S(x'), S(x)) < \rho$ return θ and stop, otherwise goto (1).

Let $p(\theta)$ be the density of a sample returned by this algorithm. Let E_x be the event $d(S(X), S(x)) < \rho$ and let $r(x) = \int \pi(\theta')(1 - P(E_x|\theta'))d\theta'$ be the probability to reject at step 2. Partitioning on the number of rejections,

$$\begin{aligned} p(\theta) &= \pi(\theta)P(E_x|\theta) + \pi(\theta)P(E_x|\theta)r(x) + \pi(\theta)P(E_x|\theta)r(x)^2 + \dots \\ &= \pi(\theta)P(E_x|\theta) \times \frac{1}{1 - r(x)} \\ &= \pi(\theta)P(E_x|\theta) \times \frac{1}{1 - (1 - \int \pi(\theta')P(E_x|\theta')d\theta')} \\ &= \pi(\theta|d(S(X), S(x)) < \rho) \end{aligned}$$

- (iii) An ABC algorithm yields samples $\theta^{(t)}, t = 1, 2, \dots, T$ with associated statistics $S^{(t)} = (S_1^{(t)}, \dots, S_p^{(t)})$. Explain how to make a regression adjustment to the sampled θ -values. What is the purpose of this adjustment?

Solution: (harder bookwork) Regress $\theta^{(t)} \sim S^{(t)}$. Suppose the estimated regression parameters are $\hat{\beta}$. The adjustment is $\theta_{\text{adj}}^{(t)} = \theta^{(t)} + (S(x) - S^{(t)})\hat{\beta}$. The adjustment shifts the distribution of θ at $S(X)$ via a linear approximation to the distribution of θ at $S(x)$ which is what we want. The purpose is to improve the approximation, allowing us to use larger tolerance ρ .

- (b) [10 marks] Now suppose $Y = y$ with $y = (y_1, \dots, y_n)$ is observed and X and θ are unknown. The Ising model is a random process for binary $m \times m$ images. For $i = 1, 2, \dots, n$ with $n = m^2$, $x_i \in \{0, 1\}$. Let N_i be the set of neighbors of pixel i on the

square lattice. Let $\#x$ denote the number of disagreeing neighbors, that is

$$\#x = \frac{1}{2} \sum_{i=1}^{m^2} \sum_{j \in N_i} \mathbb{I}_{x_i \neq x_j}.$$

Under the Ising model, $\pi(x|\theta) = \exp(-\theta\#x)/Z(\theta)$ where $Z(\theta)$ is a normalising constant. Let $y_i \sim N(x_i, \sigma^2)$ for $i = 1, 2, \dots, n$ with $n = m^2$. A prior $\theta \sim \text{Exp}(2)$ is elicited for θ .

- (i) Write down the posterior $\pi(\theta, x|y)$ in terms of the model elements and explain why it is doubly intractable when $m \gg 1$.

Solution: (variation on bookwork)

$$\pi(\theta, x|y) \propto \left[\prod_{i=1}^n N(y_i; x_i, \sigma^2) \right] \frac{e^{-\theta\#x}}{Z(\theta)} \text{Exp}(\theta; 2)$$

is DI because $Z(\theta) = \sum_{x \in \{0,1\}^n} \exp(-\theta\#x)$ is intractable.

- (ii) Consider the statistic

$$S(y) = \frac{1}{2} \sum_{i=1}^{m^2} \sum_{j \in N_i} (y_i - y_j)^2.$$

Briefly motivate this choice of the ABC statistic.

Solution: (new) When the observation noise σ is small, $S(y) \simeq \#x$ which is sufficient for θ .

- (iii) An MCMC algorithm for $X \sim \pi(x|\theta)$ is available. Give an ABC algorithm targeting $\pi(\theta|d(S(Y), S(y)) < \rho)$ using the MCMC algorithm to simulate $X \sim \pi(x|\theta)$.

Solution: (new)

(1) Simulate $\theta \sim \pi(\theta)$, $x' \sim \pi(x'|\theta)$ (using the MCMC algorithm) and $y' \sim N(y'; x', \sigma^2 I_n)$.

(2) If $d(S(y'), S(y)) < \rho$ return θ and stop, otherwise goto (1).

- (iv) Explain briefly why your algorithm targeting $\pi(\theta|d(S(Y), S(y)) < \rho)$ will not in general be exact.

Solution: (new) For any finite run length, the MCMC samples $\pi(x'|\theta')$ will not be exactly distributed according to $\pi(x'|\theta')$, they will only converge in distribution to $\pi(x'|\theta')$.

3. For $i = 1, 2, \dots, n$ let $y_i \sim f(y_i|\theta_i)$ where $\theta_i \in \Omega$. A random distribution $G \sim DP(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H on Ω . Given G , the θ_i , $i = 1, 2, \dots, n$ are independent and identically distributed according to G . Let $\theta = (\theta_1, \dots, \theta_n)$ and let $f(y|\theta) = \prod_{i=1}^n f(y_i|\theta_i)$.

(a) [2 marks] Give a definition of a Dirichlet Process.

Solution: (bookwork) $G \sim DP(\alpha, H)$ iff $\forall$ partitions $A_1, A_2, \dots, A_r$ of Ω

$$G(A_1), \dots, G(A_r) \sim \text{Dirichlet}(\alpha H(A_1), \dots, \alpha H(A_r))$$

(b) [7 marks] Show that for sets $A \subseteq \Omega$,

$$\Pr(\theta_{n+1} \in A|\theta) = \frac{\alpha H(A) + \sum_{i=1}^n \mathbb{I}_{\theta_i \in A}}{\alpha + n}.$$

You may assert without proof that if $G \sim DP(\alpha, H)$ and $(\theta_n|G) \sim G$ then $\theta_n \sim H$.

Solution: (sketched in lectures with many details missing) If $G \sim DP(\alpha, H)$ then $\theta_1 \sim H$. Next since $\Pr(\theta_1 \in A|G) = G(A)$,

$$p(G(A_1), \dots, G(A_r)|\theta_1) \propto \prod_{a=1}^r G(A_a)^{\mathbb{I}_{\theta_1 \in A_a} + \alpha H(A_a) - 1}$$

and hence

$$G(A_1), \dots, G(A_r)|\theta_1 \sim \text{Dirichlet}(\alpha H(A_1) + \mathbb{I}_{\theta_1 \in A_1}, \dots, \alpha H(A_r) + \mathbb{I}_{\theta_1 \in A_r}).$$

This is a DP if $G|\theta_1 \sim DP(\tilde{\alpha}, \tilde{H})$ and we see that if we set $\tilde{H} = (\alpha H + \mathbb{I}_{\theta_1})/\tilde{\alpha}$ we get $\tilde{\alpha}\tilde{H} = (\alpha H + \mathbb{I}_{\theta_1})$ as the power for $G(A)$ as required. We choose $\tilde{\alpha} = \alpha + 1$ so that $\tilde{H}$ is normalised, since $\alpha H(\Omega) + \mathbb{I}_{\theta_1 \in \Omega} = \alpha + 1$ and conclude that

$$G|\theta_1 \sim DP(\alpha + 1, (\alpha H + \mathbb{I}_{\theta_1})/(\alpha + 1)).$$

Now if by induction if

$$G|\theta_1, \dots, \theta_{n-1} \sim DP(\tilde{\alpha}, \tilde{H})$$

then

$$G|\theta_1, \dots, \theta_n \sim DP(\tilde{\alpha} + 1, (\tilde{\alpha}\tilde{H} + \mathbb{I}_{\theta_n})/(\tilde{\alpha} + 1))$$

so if $\tilde{\alpha} = \alpha + n - 1$ and

$$\tilde{H} = \frac{\alpha H + \sum_{i=1}^{n-1} \mathbb{I}_{\theta_i}}{\alpha + n - 1}$$

then

$$G|\theta_1, \dots, \theta_n \sim DP(\alpha + n, (\alpha H + \sum_{i=1}^n \mathbb{I}_{\theta_i})/(\alpha + n)),$$

and if $\theta_{n+1} \sim G|\theta_1, \dots, \theta_n$ then

$$\theta_{n+1}|\theta_1, \dots, \theta_n \sim \frac{\alpha H + \sum_{i=1}^n \mathbb{I}_{\theta_i}}{\alpha + n}$$

marginally.

- (c) [7 marks] Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ list the distinct values in θ , with $K \in \{1, 2, \dots, n\}$. Let $S = (S_1, \dots, S_K)$ be the partition of $\{1, 2, \dots, n\}$ which satisfies $i \in S_k$ if $\theta_i = \theta_k^*$ and let $|S_k| = \text{card}(S_k)$. Let $P(S)$ be the probability θ has partition S . Show that

$$P(S) = \frac{\alpha^K \prod_{k=1}^K (|S_k| - 1)}{\prod_{i=1}^n (\alpha + i - 1)}.$$

Solution: (fill in gaps in bookwork to do this) We note that if $S^i = (S_1^i, \dots, S_K^i)$ is the partition after i θ 's have been added then

$$\sum_{j=1}^i \mathbb{I}_{\theta_j} = \sum_{k=1}^{K^i} |S_k^i| \mathbb{I}_{\theta_k^*},$$

and so the iteration above is

$$\theta_{i+1} \sim \frac{\alpha H + \sum_{k=1}^{K^i} |S_k^i| \mathbb{I}_{\theta_k^*}}{\alpha + i}.$$

To simulate the θ 's we determine θ_{i+1} by simulating $\theta_{i+1} \sim H$ wp $\alpha/(\alpha + i)$ and setting $\theta_{i+1} = \theta_k^*$ wp $|S_k^i|/(\alpha + i)$. Consider the sequence of independent events leading to a particular partition $S = (S_1, \dots, S_K)$. The first θ is in a new cluster with probability one. If we end up with K clusters then the process must have simulated $K - 1$ new θ 's, and each of these events has probability proportional to α , giving an overall factor α^{K-1} . The k 'th cluster filled up to $|S_k|$ elements with events with probabilities proportional to $1, 2, \dots, |S_k^i| - 1, \dots, |S_k| - 1$, giving a factor $(|S_k| - 1)!$. The $i + 1$ st event had a denominator $\alpha + i$ so the last event has a denominator $\alpha + n - 1$. The first event (determining θ_2) has denominator $\alpha + 1$, so the denominator is $\prod_{i=2}^n (\alpha + i - 1)$. We get expression above starting the product at $i = 1$ and adding a factor α in the numerator.

- (d) [7 marks] Suppose that for $k = 1, 2, \dots$, $H(d\theta_k^*) = h(\theta_k^*)d\theta_k^*$. Let $h(\theta^*) = \prod_{k=1}^K h(\theta_k^*)$ and let $S^{-i} = (S_1^{-i}, \dots, S_K^{-i})$ denote the partition S with i removed. Explain why

$$\pi(d\theta|y) \propto f(y|\theta^*, S)h(d\theta^*)P(S)$$

and show that

$$\Pr(i \in S_k | S^{-i}, \theta^*) = \begin{cases} |S_k^{-i}| f(y_i | \theta_k^*) & \text{if } k \in \{1, 2, \dots, K\} \\ \alpha f(y_i | \theta_k^*) & \text{if } k = K + 1. \end{cases}$$

Solution: (many new elements) The first part is just Bayes rule, noting that θ and S, θ^* are in one to one correspondence, and also that, given the partition S , $\theta_i = \theta_k^*$ for each $i \in S_k$ with $\theta_k^* \sim H$ iid for $k = 1, 2, \dots, K$ (from the iterative procedure generating the θ 's). Then, $\pi(d\theta|y) \propto f(y|\theta)\pi(d\theta)$ with $\pi(d\theta) = \pi(d\theta^*, S) = h(d\theta^*)P(S)$.

For the conditional distributions we need

$$\pi(d\theta^*, S^{-i} | y^{-i}) \propto \left[\prod_{j \neq i} f(y_j | \theta_j) \right] \left[\prod_{k=1}^{K+1} h(d\theta_k^*) \right] \left[\alpha^K \prod_{k=1}^K (|S_k^{-i}| - 1)! \right],$$

and notice we have added an empty cluster, with an associated θ^* -value. This cluster wasn't occupied in the θ -prior simulation so there is no factor $\alpha(|S_{K+1}^{-i}| - 1)!$. If we integrate over the values of θ_{K+1}^* we simply remove the factor $\int h(d\theta_{K+1}^*) = 1$ so this is the same distribution. There are an infinite number of empty clusters each with its own θ^* value, and we have marginalised over those θ^* values. We add a generic empty cluster to the model to avoid the dimension of θ^* changing if we put i into a new cluster, ie $\pi(d\theta^*, S^{-i}|y^{-i})$ and $\pi(d\theta^*, S|y)$ have densities with respect to the same measure $d\theta^*$.

Now

$$\Pr(i \in S_k | S^{-i}, \theta^*) = \frac{\pi(\theta^*, S|y)}{\pi(\theta^*, S^{-i}|y^{-i})}$$

and we see that $\left[\prod_{k=1}^{K+1} h(d\theta_k^*) \right]$. We always get an extra factor $f(y_i|\theta_k^*)$, and if we put i in cluster $k = 1, 2, \dots, K$ then $(|S_k^{-i} \cup \{i\}| - 1)! / (|S_k^{-i}| - 1)! = |S_k^{-i}|$ and α^K cancels. If i goes in the empty cluster then $K \rightarrow K + 1$ so we get a factor α but $(|S_{K+1}| - 1)! = 1$ and the other clusters are unchanged, so the factorials cancel. This gives the stated conditionals.

- (e) [2 marks] Suppose $h(\theta)$ is some general (ie, non-conjugate) prior on Ω . Briefly outline an MCMC algorithm targeting $\pi(\theta|y)$.

Solution: (new - mentioned in passing in lectures) The conditionals are what we need for a Gibbs sampler updating the partition S identifying data points y_i with clusters. The update on the θ^* -values may be carried out using for example standard random walk MCMC acting on each of the θ_k^* in turn.